26021 Kemi (Polyteknisk grundlag)/26021 Chemistry (Polytechnic Foundation) non-random version

Der er kun et korrekt svar for hvert spørgsmål.
Du kan kun vælge ét svar per spørgsmål
Alle spørgsmål indgår med samme vægt i bedømmelsen.

There is only one correct answer for each question.
You can only select one answer per question.
All questions have equal weight in the evaluation.

Hvad er antallet af neutroner i ^{238}U?

What is the number of neutrons in ^{238}U?

Choose one answer

- ○ 92
- x 146
- ○ 238
- ○ 0
- ○ 119

Hvad er den kemiske formel for ammoniumnitrat?

What is the chemical formula for ammonium nitrate?

Choose one answer

- ○ $Am(NO_3)_4$
- ○ NH_4NO_3
- ○ $(NH_4)_3N$
- X NH_4NO_2
- ○ $Am(NO_3)_3$

Hvad er oxidationstallet for Cr i $Cr_2O_7^{2-}$?

What is the oxidation state for Cr in $Cr_2O_7^{2-}$*?*

Choose one answer

- ○ 0
- ○ 3
- ○ 7
- x 6
- ○ 12

Når den følgende reaktion er afstemt med det numerisk mindst mulige set af heltal, hvad er summen af koefficienterne (glem ikke koefficienter af 1):

$Cr_2O_7^{2-} + H_2S \longrightarrow Cr^{3+} + S$ (sur opløsning)

When the following reaction is balanced with the smallest possible set of integers, what is the sum of all the coefficients (do not forget coefficients of one)

$Cr_2O_7^{2-} + H_2S \longrightarrow Cr^{3+} + S$ (acidic solution)

Choose one answer

- ○ 29
- ○ 13
- X 24
- ○ 19
- ○ 7

Hvad er massen af 1 mmol (millimol) ammoniumhydrogenphosphat?

What is the mass of 1 mmol (millimol) ammonium hydrogenphosphate?

Choose one answer

- ○ 115 g
- ○ 132 g
- ○ 1.15 g
- x 0.132 g
- ○ $6.02 \cdot 10^{-20}$ g

Hvilken af følgende har den største masse?

Which of the following have the largest mass?

Choose one answer

x 1 mol $SF_6(g)$

- 1 mol $Fe(s)$
- 1 mol $CO_2(g)$
- De har alle samme masse [*They all have the same mass*]
- 1 mol $CH_3CH_2OH(l)$

Hvad er massen af dihydrogen der dannes når aluminium (25 g) reagerer med saltsyre ifølge denne (ikke afstemte) reaktion:

$Al + HCl(aq) \longrightarrow AlCl_3 + H_2$

What is the mass of dihydrogen formed when aluminium (25 g) reacts with hydrochloric acid according the (unbalanced) reaction:

$Al + HCl(aq) \longrightarrow AlCl_3 + H_2$

Choose one answer

- X 2.8 g
- ○ 0.41 g
- ○ 1.9 g
- ○ 0.93 g
- ○ 1.2 g

QUESTION 8

En koncentrationscelle er konstrueret ved at placere identiske kobberelektroder i two kobber(II)-opløsninger. De to opløsninger er 1.0 M og 2.0 mM i Cu^{2+}-koncentration. Udregn den elektromotoriske kraft for den spontant forløbende process i koncentrationscellen, E_{cell}.

A concentration cell is constructed by placing identical copper electrodes in two copper(II) solutions. The two solutions are 1.0 M and 2 mM in Cu^{2+} concentration. Calculate the electromotive force of the spontaneous process in the concentration cell, E_{cell}.

Choose one answer

- ○ -0.080 V
- X +0.080 V
- ○ +0.020 V
- ○ -0.34 V
- ○ 0.34 V

Hvilken af følgende er den stærkeste syre?

Which of the following is the strongest acid?

Choose one answer

- X $H_2SO_4(aq)$
- ○ $H_2O(l)$
- ○ $H_3PO_4(aq)$
- ○ $H_2PO_4^{-}(aq)$
- ○ $HPO_4^{2-}(aq)$

Betragt en given kemisk reaktion og dets ligevægtskonstant, *K*. Hvilket af følgende udsagn er korrekt?

Consider a given chemical reaction and its equilibrium constant, K. Which of the following statements is correct?

Choose one answer

x *K* ændres hvis temperaturen ændres [*K changes if the temperature changes*]

○ *K* forbliver den samme hvis temperaturen ændres [*K remains the same if the temperature changes*]

○ *K* formindskes hvis koncentrationen af en reaktant formindskes [*K decreases if the concentration of a reactant is decreased*]

○ *K* forøges hvis koncentrationen af en af produkterne forøges [*K increases if the concentration of one of the products is increased*]

○ *K* ændres hvis en katalysator bliver tilført [*K changes by if a catalyst is added*]

Hvilket volumen NaOH(aq) (0.5 M) vil du have brug for, for at omdanne en 500 mL-opløsning af H_3PO_4 (0.25 M) til en opløsning af natriumphosphat?

What volume of NaOH(aq) (0.50 M) would you need to convert a 500 mL-solution of H_3PO_4 (0.25 M) into a solution of sodium phosphate?

Choose one answer

- X 0.75 L
- ○ $2.5 \cdot 10^2$ mL
- ○ $1.4 \cdot 10^2$ mL
- ○ 83 mL
- ○ 1.5 L

Betragt en opløsning som er 0.10 M i CH_3COOH og 0.20 M i $NaCH_3COO$. Hvilket af følgende udsagn er korrekt?

Consider a solution which is 0.10 M in CH_3COOH and 0.20 M in $NaCH_3COO$. Which of the following statements is true?

Choose one answer

- ○ Hvis en lille mængde OH^- tilsættes, formindskes pH ganske lidt. [*If a small amount of OH^- is added, the pH decreases very slightly*]
- ○ Hvis natriumhydroxid tilsættes vil OH^- reagere med CH_3COO^-. [*If sodium hydroxide is added, OH^- reacts with CH_3COO^-*]
- x Hvis en lille mængde HCl(aq) tilsættes, formindskes pH meget lidt. [*If a small amount of HCl(aq) is added, the pH decreases very slightly*]
- ○ Hvis saltsyre tilsættes reagerer H^+ med CH_3COOH. [*If hydrochloric acid is added, the H^+ ions react with CH_3COOH*]
- ○ Hvis der tilsættes mere CH_3COOH, vil pH forøges. [*If more CH_3COOH is added, the pH increases*]

Tre flasker (2 L) fyldes respektivt med H_2, O_2 og He, ved $T = 25$ °C and $P = 1$ bar. Hvilket af følgende udsagn er korrekt? Gasserne kan betragtes som værende ideale.

Three flasks (2 L) are filled with H_2, O_2, and He, respectively, at $T = 25$ °C and $P = 1$ bar. Which of the following statements is correct? The gases can be considered as ideal.

Choose one answer

- ○ Molekylernes hastighed er identisk i alle flaskerne. [*The velocity of the gas molecules is identical in each flask*]
- X Antallet af O_2-molekyler i den ene flaske er identisk med antallet af He-atomer i den anden. [*The number of O_2 molecules in one flask is identical to the number of He atoms in the other flask*]
- ○ Densiteten af gasserne er den samme. [*The density of each gas is the same*]
- ○ Der er dobbelt så mange H_2-molekyler som der er He-atomer. [*There are twice as many H_2 molecules as there are He atoms*]
- ○ Ingen af disse er korrekte. [*None of these are correct*]

QUESTION 14

De følgende opløselighedsprodukter for chromater (CrO_4^{2-}-salte) er blevet bestemt (ved 25 °C):

	K_{sp}
Ag_2CrO_4	$9.0 \cdot 10^{-12}$ M^3
$BaCrO_4$	$2.0 \cdot 10^{-10}$ M^2
$PbCrO_4$	$1.8 \cdot 10^{-14}$ M^2

Chromatet der er **mest** opløseligt (i mol/L) ved 25 °C er:

The following solubility products for three chromates (CrO_4^{2-} salts) have been determined (at 25 °C):

	K_{sp}
Ag_2CrO_4	*$9.0 \cdot 10^{-12}$ M^3*
$BaCrO_4$	*$2.0 \cdot 10^{-10}$ M^2*
$PbCrO_4$	*$1.8 \cdot 10^{-14}$ M^2*

*The chromate that is the **most** soluble (in mol/L) at 25 °C is:*

Choose one answer

- ○ $BaCrO_4$
- ⊗ Ag_2CrO_4
- ○ Alle er fuldt opløselige i vand. [*All are fully soluble in water*]
- ○ $PbCrO_4$
- ○ Umuligt at bestemme uden yderligere information. [*Impossible to determine without additional information*]

Atomet med elektronkonfigurationen $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$ vil være i:

The atom with the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$ would be in:

Choose one answer

- ◯ 6. hovedgruppe, 5. periode [*Main group 6, period 5*]
- ⊗ 7. hovedgruppe, 4. periode [*Main group 7, period 4*]
- ◯ 4. hovedgruppe, 4. periode [*Main group 4, period 4*]
- ◯ 4. hovedgruppe, 7. periode [*Main group 4, period 7*]
- ◯ 6. hovedgruppe, 5. periode [*Main group 6, period 5*]

Betragt følgende reaktion:

$$SBr_4(g) \longrightarrow S(g) + 2\,Br_2(l)$$

ved $T = 25$ °C, $\Delta H° = +115$ kJ mol^{-1} og $\Delta S° = +125$ J K^{-1} mol^{-1}. Udregn $\Delta G°$ for reaktionen ved 25 °C.

Consider the following reaction:

$$SBr_4(g) \longrightarrow S(g) + 2\,Br_2(l)$$

at $T = 25$ °C, $\Delta H° = +115$ kJ mol^{-1} and $\Delta S° = +125$ J K^{-1} mol^{-1}. Calculate $\Delta G°$ for the reaction at 25 °C.

Choose one answer

- ○ –56.7 kJ mol^{-1}
- ⓧ +77.8 kJ mol^{-1}
- ○ –86.2 kJ mol^{-1}
- ○ +152 kJ mol^{-1}
- ○ +112 kJ mol^{-1}

Betragt igen reaktionen:

$$SBr_4(g) \longrightarrow S(g) + 2\ Br_2(l)$$

ved $T = 25$ °C, $\Delta H° = +115$ kJ mol^{-1} og $\Delta S° = +125$ J K^{-1} mol^{-1} (som begge kan antages som værende temperaturuafhængige). Denne reaktion er:

Consider again the reaction:

$$SBr_4(g) \longrightarrow S(g) + 2\ Br_2(l)$$

at $T = 25$ °C, $\Delta H° = +115$ kJ mol^{-1} and $\Delta S° = +125$ J K^{-1} mol^{-1} (both of which can be assumed independent of temperature). This reaction is:

Choose one answer

- (x) Spontan ved temperaturer over ca. 900 K. [*Spontaneous at temperatures above ca. 900 K*]
- ( ) Spontan ved alle temperaturer. [*Spontaneous at all temperatures*]
- ( ) Spontan ved temperaturer under ca. 900 K. [*Spontaneous at temperatures below ca. 900 K*]
- ( ) Kun spontan ved 25 °C. [*Only spontaneous at 25 °C*]
- ( ) Ikke-spontan ved alle temperaturer. [*Non-spontaneous at all temperatures*]

De fleste kemiske reaktioner forløber hurtigere ved højere temperaturer end ved lavere temperaturer. Dette skyldes:

(I) en forøgelse i reaktionens aktiveringsenergi, E_a, med stigende temperatur.
(II) en forøgelse af reaktionens hastighedskonstant, k, med stigende temperatur.
(III) en forøgelse i molekylers gennemsnitlige hastighed med stigende temperatur.

The majority of chemical reactions are more rapid at higher temperatures than at lower temperatures. This observation is due to:

(I) an increase in the activation energy, E_a, of the reaction with increasing temperature.
(II) an increase in the rate constant, k, with increasing temperature.
(III) an increase in the average velocity of the molecules with increasing temperature.

Choose one answer

- ○ Kun (I). [*Only (I)*]
- ○ Kun (II). [*Only (II)*]
- ○ Både (I), (II) og (III). [*All of (I), (II), and (III)*]
- ⊗ Kun (II) og (III). [*Only (II) and (III)*]
- ○ Kun (I) og (II). [*Only (I) and (II)*]

Hvilke(t) af følgende fire molekyler er polært: PH_3, H_2O, HCl, SO_3?

Which of the following more molecules is/are polar: PH_3, H_2O, HCl, SO_3?

Choose one answer

- ○ Kun HCl og H_2O. [*Only HCl and H_2O*]
- ○ Kun HCl. [*Only HCl*]
- ⊗ Alle undtagen SO_3. [All except SO_3]
- ○ Ingen af dem. [*None of them*]
- ○ Allesammen. [*All of them*]

QUESTION 20

For grundstofferne i 2. hovedgruppe, forøges alle de følgende når man går ned igennem gruppen, pånær:

For the main group 2 elements, all of the following increase with going down the group except:

Choose one answer

- ○ Atomets volumen. [*The volume of the atom*]
- (x) Den første ioniseringsenergi. [*The first ionization energy*]
- ○ Den molare masse. [*The molar mass*]
- ○ Ionradius. [*The ionic radius*]
- ○ Atomradius. [*The atomic radius*]